Investigating the Impact of Bill 187 on Ontario's Electronics Industry

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Abstract

This paper examines the potential economic implications of Bill 187, which aims to improve consumer rights to repair products, on the electronics industry in Ontario. Through a qualitative analysis of barriers, a comparative analysis of international right-to-repair legislation, and descriptive analysis of Ontario's electronics manufacturing and repair industries, the paper evaluates whether Bill 187 addresses long-standing obstacles to repair. The findings show that while Bill 187 improves access to spare parts and information, it leaves major competitive and consumer behavioural barriers unaddressed, including planned obsolescence, pricing, and convenience and trust. Comparisons with right-to-repair legislation in the EU, UK, and the United States reveal that Bill 187 is less comprehensive in its treatment of long-term parts availability and anti-obsolescence measures. Despite these limitations, Bill 187 has the potential to improve conditions for small Ontario manufacturers, stimulate Ontario's repair sector and improve consumer access to repair.

1 Introduction

Repairs for damaged electronics have become increasingly difficult for Canadians in recent years as devices have become smaller and more complex in nature. According to a 2021 national survey, just 18% of Canadians had their most recently broken electronic or home appliance repaired (Équiterre 2022, 9). An overwhelming majority of consumers are choosing to replace damaged or dead devices instead of repairing them because the costs of the two options are often similar, if repair is even an option.

Unlike the past, the repair work of today requires more than analyzing circuits and replacing components. Manufacturers of cell phones, home appliances and other consumer electronics have implemented legal, digital and physical barriers that prevent consumers from doing their own repairs or using independent repair shops (Hessey 2023). Increasingly, electronics are made with proprietary parts controlled exclusively by manufacturers through copyright, trademarks and patents. When these parts are not made available to third parties, repair becomes very difficult, if not impossible.

Planned obsolescence is a term used to describe the strategy of intentionally designing products to have a shorter lifespan and to become obsolete after a certain amount of time (López-Bermúdez et al. 2025, 3). A frequently cited example of planned obsolescence in the electronics industry is Apple's 2017 scandal over iPhone processor slowdowns. 'Batterygate' was termed to describe Apple's intentional slowing of older iPhone models through software updates. The result was a huge loss in consumer trust, numerous class action lawsuits and fines ranging from €25 million in France (Burnett 2020) to $500 million in California (Robertson 2020) and $14.4 million CAD in Canada (Epiq Class Action Services Canada 2025).

Aside from the financial costs of replacing a damaged electronic, there are also environmental costs associated with the constant replacement of goods. The 2019 Economic Study of the Canadian Plastic Industry, Markets and Waste, commissioned by Environment and Climate Change Canada, found that electrical and electronic equipment (EEE) is one of the main generating sectors for plastic waste, with 87% of plastic waste ending up in landfills or the environment (Environment and Climate Change Canada 2019). This poses a threat to human health and the environment.

The Government of Canada has established a Strategy on Zero Plastic Waste, committing to taking a circular economy and lifecycle approach to plastics. Remanufacturing, refurbishment, repair and direct reuse (RRRDR) is the first option in the waste management hierarchy, but the least prevalent in Canada (Environment and Climate Change Canada 2019). RRRDR is vital, as durable goods account for 51% of total plastics waste (Environment and Climate Change Canada 2019). Initiatives to repair electronic products exist, but remain fragmented and small-scale in nature, and several factors limit the development of RRRDR approaches, such as a negative tradeoff for the consumer, product irreparability by design, and the dominance of linear business models (Environment and Climate Change Canada 2019).

In Canada, there has been growing support amongst consumers of "Right to Repair" (RTR) legislation. A national survey conducted in 2019 found that regardless of political party identification, age, gender or region, 75% of Canadians would support RTR legislation (Innovative Research Group 2019). The RTR movement calls for the government to pass legislation that gives consumers greater freedom to repair the products they own, rather than being forced to use the manufacturer's service or buy a new one. Advocates of the RTR

movement cite greater choice, consumer autonomy and reduced waste as positive outcomes (Nightingale 2024).

Globally, the RTR movement has made a lot of progress. In the United States, RTR bills have gained increasing momentum in recent years, with states like New York, Colorado, Minnesota, California, and Oregon all having passed RTR bills on electronics in the last few years (PIRG 2025). Just last July in 2024, the EU officially enacted its Right-to-Repair Law, mandating that all manufacturers make repair services more accessible, transparent, and attractive (Garden 2024).

Following a public consultation in 2024, the Canadian Government has been taking steps towards implementing RTR policies for consumers. Canada's Parliament enacted Bills C-244 and C-294 in late 2024, which amended the Copyright Act to allow a person to circumvent a technological protection measure (TPM) to diagnose, maintain or repair certain types of devices (Chau 2024). Currently, the Ontario legislature has taken the first steps to pass RTR legislation, introducing Bill 187, which passed its first reading in April 2024. Bill 187 proposes to amend the Consumer Protection Act to require manufacturers to provide repair manuals, replacement parts and software and tools to allow consumers and businesses to diagnose, maintain, or repair electronic products, household appliances, wheelchairs, motor vehicles and farming heavy equipment (Legislative Assembly of Ontario 2024). The manufacturer would also be required to provide the repair manual at no charge, and the replacement parts, software and tools at a fair cost.

There is limited research on how specific RTR policies such as Bill 187 will impact local market participants and the surrounding environment. This paper seeks to examine the economic

and environmental implications for Ontario if Bill 187 is adopted, focusing on the electronics sector. Some of the research questions that will be examined in this paper include:

- Does Bill 187 overcome the typical barriers that RTR policies face?
- How does Bill 187 compare with similar international legislation?
- How will manufacturers and consumers in Ontario be affected?
- What is the potential impact on electronic waste reduction in Ontario?

2 Literature Review

There has been considerable research on many of the topics related to this paper. Analysis of the history of the right to repair in Canada has found that potential obstacles in realizing right-to-repair reforms include constitutional restrictions on Parliament's legislative power and the need to find public support for the right to repair (Rosborough 2022, 104). Alongside stronger federal-provincial cooperation, equipping consumers with information to make informed decisions would be required, such as through the creation of a federal sustainability index with repairability scores (Rosborough 2022, 123).

Right to repair has been identified as a key driver in the transition from a linear supply chain to a circular supply chain through remanufacturing for electrical and electronic equipment (Werner-Lewandowska et al. 2025, 14). In turn, circular supply chains are key to developing a circular economy. Electrical and electronic equipment has been identified as one of the priority areas for a circular supply chain implementation due to its high potential for circularity (Werner-Lewandowska et al. 2025, 1).

Jin, Yang and Zhu (2022) examine how legislation would shape manufacturers' decisions in the product market as they try to mitigate profit losses. A study of the pricing, welfare, and environmental implications of RTR has found that when production costs are low, both consumer

surplus and social welfare improve, while environmental impact increases as the manufacturer adopts a volume strategy through lowering the new product price (Jin et al. 2022, 1017). When the production cost is intermediate, consumer surplus will first increase along with a price drop, then decrease along with a price hike, and so will social welfare (Jin et al. 2022, 4). The total environmental impact will follow a similar pattern if the production and disposal phases are the main causes of impact; however, if the use phase is the main contributor to environmental impact, the total impact will continue to increase as the independent repair cost falls (Jin et al. 2022, 4). This implies that RTR can lead to a win-lose, a lose-win, or a lose-lose outcome for consumers and the environment, depending on the market conditions.

Chen and Morteza (2023) have examined the impact of RTR policies on sales, repair demand, manufacturer profitability, consumer surplus, the environment and social welfare. They found that optimal policies differ depending on each product's repairability. In certain situations, a repair bonus policy is more effective than a policy mandating that manufacturers provide repair resources to consumers. Subsidy policies are better for products that are easy to repair, while policies mandating the availability of repair resources are more suitable for products that are harder to repair (Chen and Morteza 2023, 29). Regulating manufacturers to reduce the use-phase impact of their products would also enhance environmental protection measures (Chen and Morteza 2023, 29).

Outside of the electronics sector, the synthetic control method estimation of the impact of Massachusetts' 2012 Right to Repair law for automakers has shown that right-to-repair laws can lead to an increase in the number and market share of small independent repair shops (Kahane 2021, 873). Across international examples, including the EU and US, research finds that even when products are designed to last, social factors such as perceived technical obsolescence,

socioeconomic position, and surface degradation still drive buyers to keep purchasing the newest and best products (Kalnawat and Nuzhat 2022, 52). In this view, law by itself is insufficient, and legislation needs to support several strategies, including educating consumers about planned obsolescence, repairing and recycling and the circular economy.

Marikyan and Papagiannidis (2024) applied a stimulus-organism-response model on US data to find that consumers' intention to repair is dependent on three groups of factors: pro-environmental behaviour, repair-related factors and beliefs about legislation (Marikyan and Papagiannidis 2024, 35). Looking into the importance of consumer barriers to repair (Roskladka et al. 2023, 1), it has been found that convenience of repair and the technical possibility of repair are the two most significant barriers. Consumer practices and routines have an important role, as do the types of products (Jaeger-Erben et al. 2020, 1).

Habib, Mohammadi and Withanage (2023) provide a detailed analysis of estimates of e-waste in Canada. Using dynamic Material Flow Analysis and trade data, the in-use stocks of EEE and e-waste generation have been calculated in Canada from 1971 to 2030 for 51 product categories. E-waste generation per capita has tripled over the past two decades, from 8.3kg in 2000 to 25.3 kg in 2020, and is estimated to reach 31.5kg by 2030 (Habib et al. 2023, 1). Due to the presence of a large number of hazardous materials such as lead, mercury, cadmium, and arsenic, e-waste is generally considered a hazardous waste. If improperly managed, it can cause significant environmental impacts and human health implications.

Lastly, there has been research analyzing the potential impact of new right-to-repair rules on electrical and electronic equipment waste in the UK (Nagase and Uehara 2024, 175). The findings of this research show that longer EEE use and better e-waste collection appear to be effective in reducing landfilled e-waste, while more reuse, recycling and recovery have

negligible impacts. It was found that one year longer EEE use and 10% more of e-waste collection could at maximum reduce landfilled e-waste by 14.05% for monitoring and control instruments and 93.93% for display equipment (Nagase and Uehara 2024, 182).

Overall, these works provide a consistent analytical framework for evaluating regulations on the right-to-repair in other jurisdictions, but do not explicitly consider the local environment of Ontario or the provincial impacts that potential right-to-repair legislation like Bill 187 will have. My research aims to fill this gap by assessing Bill 187's efficacy with respect to common barriers to RTR legislation, performing a comparative analysis of Bill 187 to similar international RTR legislation, analyzing the makeup of Ontario's electronics industry, and finally examining the potential for e-waste reduction in Ontario.

3 Methods

This paper evaluates the potential impacts of Ontario's proposed Bill 187 on the province's electronics manufacturing industry, repair sector as well as consumers. The methodology combines qualitative policy analysis, comparative analysis of international legislation, and descriptive analysis of industry data. These approaches are adapted from those used in recent right-to-repair research, particularly Lopez-Bermudez & Vence (2025) and Jin et al. (2020), applying their models to the Ontario context.

3.1 Framework for Identifying Barriers to Repair

To assess whether Bill 187 addresses the barriers commonly found to impede repair, the barrier classification framework created by Lopez-Bermudez & Vence (2025) is used. Their work categorizes obstacles into access barriers (legal, spare parts, information), competitive barriers (pricing and planned obsolescence), and consumer-level barriers. Using the table provided in

their methodology, each barrier is evaluated against the specific provisions of Bill 187. The result is a structured determination of whether Bill 187 fully addresses a barrier, partially addresses it, or leaves it unaddressed.

3.2 Comparative Analysis of International Regulation

To compare Bill 187 against international right-to-repair regulation, a comparative analysis is performed of major RTR laws in the European Union, the United Kingdom and various U.S. states.

Key aspects that were compared include:

- The scope of covered products
- Mandatory availability of spare parts and tools
- Timeframes for parts availability
- Price standards for parts and manuals
- Provisions targeting planned obsolescence
- Repairability indexes or consumer information tools

Through this comparative analysis, it is assessed whether Bill 187 is more or less comprehensive than existing international legislation, and to identify if there are features that are not currently included in the bill.

3.3 Industry Data Collection and Descriptive Analysis

To understand how Ontario's manufacturers and repair businesses may respond to Bill 187, this paper conducts a descriptive analysis of industry structure and trade flows in the electronics industry. Industry data for 2024 is collected from Canadian Industry Statistics for NAICS Group 334 (Computer and electronic product manufacturing) and Group 811 (Repair and maintenance), including:

- Number of businesses by province and firm size
- Subsector composition
- Financial performance of businesses by quartile

After analyzing the electronics manufacturing industry structure, the financial performance of manufacturing businesses is used to identify the relative production costs between subsectors.

Jin et. al (2020) analyzed the pricing and welfare implications of RTR legislation on pricing and welfare. In the model, the three cases revolve around the manufacturer's unit production cost c and how that compares to a lower cutoff $\underline{c}$ and an upper cutoff $\bar{c}$. More information regarding each category is outlined in Table 3.

- Low production cost: $c \leq \underline{c}$
- Intermediate production cost: $\underline{c} \leq c < \bar{c}$
- High production cost: $c \geq \bar{c}$

Since no publicly available data exists that could be used to estimate the lower and upper cutoffs required by the theoretical framework provided, this paper instead uses data from Canadian Industry Statistics to classify manufacturers. Gross margin is used as a proxy for production costs, because it directly reflects how much revenue remains after accounting for the cost of goods sold. Industries with higher gross margins typically have lower marginal production costs and higher pricing power (Vance 2021). On the other hand, low gross margins indicate higher unit production costs and less flexibility to absorb pricing pressures. Thus, gross margin provides a practical and consistent way to map manufacturers into the "low-cost," "intermediate cost," and "high cost" production categories outlined by Jin et al. (2022).

This model is used as a reference for grouping electronics manufacturers into low-cost, intermediate-cost and high-cost producers, based on their gross margins, which are used as a

proxy for underlying costs of production, relative to the output price. To evaluate the relative competitiveness of domestic producers and the degree to which Ontario relies on imported electronics, 2024 electronics manufacturing trade data is taken from Trade Data Online.

Survey data from Équiterre (2022) and Innovative Research Group (2019) is used to analyze consumer attitudes toward the repair sector. Findings from the literature are used to identify behavioural barriers and evaluate whether Bill 187 addresses the main consumer-level barriers to repair.

4 Results

The results are organized into four components: an evaluation of Bill 187 against established repair barriers, a comparison with international legislation, an assessment of impacts on Ontario's electronics sector and consumers, and a review of potential environmental implications.

4.1 Assessment of Bill 187 in Relation to Identified Barriers to Repair

Following a similar methodology as Lopez-Bermudez and Vence (2025), the relevance and potential impacts of the measures proposed in Ontario's Bill 187 was assessed based on the existing literature on the promotion of repair. An overview of barriers and the main obstacles to repair identified by Lopez-Bermudez and Vence (2025) is outlined in Table 1, and the corresponding evaluation of Bill 187 is presented in Table 2. In addition to access barriers and competitive barriers, Svensson et al. (2018) identify a third category of obstacles to repair, consumer preferences and attitudes. Overall, Bill 187 was found to address only a subset of known repair barriers, leaving most competitive and consumer-level barriers unaddressed.

Legal obstacles – Not addressed

Ontario's Extended Producer Responsibility program and Ontario Regulation 522/20 focus on collection and recycling of e-waste rather than extending product life through repair. Under the Consumer Protection Act (2002), consumers have implied warranties, but these do not require repair to be offered or prioritize repair over replacement.

Spare parts – Addressed

Bill 187 mandates that manufacturers make spare parts available for a minimum of 5 years after the last product is sold. This provision directly tackles one of the main access barriers identified in the literature by reducing dependence on authorized repair services and extending the usable lifespan of products.

Information – Addressed

Bill 187 requires manufacturers to provide repair manuals and diagnostic tools to consumers and independent repair providers at a reasonable cost. This reduces information asymmetries and challenges related to intellectual property restrictions which previously limited consumers' ability to repair their own products.

Prices – Partially addressed

By improving access to spare parts and repair information, Bill 187 may indirectly reduce repair costs. However, it does not include measures to make repair services more financially attractive than replacement, such as subsidies, tax incentives, or pricing regulations. Consequently, while the relative cost of repair may decrease, replacement may still be cheaper for many consumers.

Material Obsolescence – Not addressed

Bill 187 does not contain provisions that regulate material quality, durability, or design-for-repair standards. Manufacturers may still design products using fragile or fused components that are difficult or costly to repair. As a result, material obsolescence remains a major unaddressed barrier, especially in electronics and appliances.

Functional Obsolescence – Not addressed

Bill 187 does not address software compatibility or product updates that cause older devices to become obsolete. For example, products may still lose functionality due to discontinued software support or incompatible updates. This limits the long-term effectiveness of Bill 187 for digital devices and other smart technologies.

Psychological Obsolescence – Not addressed

Bill 187 focuses primarily on technical and regulatory access to repair but does not address social or behavioural drivers of replacements, such as marketing practices or consumer perceptions of "newness." Without measures promoting repair culture or awareness campaigns, psychological obsolescence will continue to be an issue.

Convenience and trust – Not addressed

Bill 187 does not contain provisions to increase consumer trust in independent repairers or to certify repair quality. The lack of standardized certification or consumer protection measures for repaired products may discourage consumers from choosing repair services, especially for high-value or safety sensitive items.

4.2 International Comparison

There have been several international approaches to right-to-repair legislation, with varying scope, enforcement mechanisms and degrees of effectiveness. Ontario's Bill 187 aligns with the global trend toward expanding consumer access to repair resources but remains less comprehensive than comparable laws in the EU, UK, and select U.S. states.

In the European Union, the 2024 Directive on the Repair of Goods and accompanying Ecodesign regulations require manufacturers to make spare parts and repair information available for up to 10 years after purchase and to prompt repair beyond the legal guarantee period. The EU also targets software-related barriers and introduces repairability scoring systems to inform consumers (European Commission 2024). In particular, France has adopted a repairability index, and has announced plans to replace it with a "Durability Index" which will address lifecycle aspects of products (European Environment Agency 2024).

The United Kingdom similarly mandates that manufacturers of in-scope appliances (e.g. washing machines, dishwashers, refrigerators, displays) provide spare parts and repair manuals for professional repairs for 7-10 years after a product's end of sale. However, access for individual consumers remains limited. In the United States, several states, including New York, California, Minnesota, and Oregon, have enacted RTR laws requiring manufacturers to provide parts, tools, and documentation on fair and reasonable terms. California's law specifies timeframes (3 years for products priced $50-$99.99, 7 years for $100+), while Oregon bans "parts pairing" practices that block independent repairs through software restrictions.

Overall, while Bill 187 addresses core access barriers by requiring manufacturers to make spare parts, tools and manuals available, it does not establish explicit timeframes for parts availability, set pricing standards, or offer protection against planned obsolescence.

Comparatively, international approaches such as the EU's 2024 Directive and recent U.S. state laws impose clearer timelines and stronger protections, though their relatively recent implementations means that their ultimate effectiveness remains unclear.

4.3 Effect on Ontarian Manufacturers, Repair Businesses and Consumers

4.3.1 Overview of Computer and Electronic Product Manufacturing Industry

Canada's computer and electronic product manufacturing sector is primarily made up of businesses that manufacture computers, computer peripheral equipment, communications equipment and similar electronic products, as well as components for these products. Across provinces, the sector experienced highly uneven growth between 2022 and 2023, ranging from a 30.2% increase in provincial GDP in Ontario to a 11.6% decrease in Yukon. This emphasizes Ontario's role as the country's central province for electronics manufacturing activities.

In 2024, there were 2,819 businesses in this sector nationally, with Ontario alone accounting for 47.4% of all employer establishments. The provincial breakdown shows that the industry is dominated by smaller firms: 57.8% of Ontario's employer establishments are small businesses, 33.4% are micro and only 0.5% of businesses qualify as large (500+ employees). This size distribution suggests that policy changes such as Bill 187 may have uneven impacts, as smaller manufacturers generally have fewer resources to absorb compliance costs.

The structure of the sector also reveals substantial variation across its six subsectors. As shown in Figures 1 and 2, navigational, measuring, medical and control instruments manufacturing represents the largest subsector, with 960 businesses in 2024 and some of the highest average revenues, alongside semiconductor and other electronic component manufacturers.

Table 4 shows the total revenue, cost of sales, net profit/loss and gross margin for each subsector, broken down into quartiles. The quartiles divide the businesses within each subsector into four groups of the same size based on levels of revenue generated. The bottom quartile consists of the 25% of businesses with the lowest reported operating revenue, the lower middle includes those above the bottom quartile cutoff but below the median. The upper middle includes those above the median but below the cut-off for the top quartile, and the top quartile consists of the 25% of businesses with the highest operating revenue.

To categorize manufacturers into cost groups, the gross margin for every quartile of every subsector was sorted from least to greatest and split into tertiles. As shown in Table 5, the cut-off points were found to be at 53.1% and 68.1% gross margin. Firms below 53.1% were classified as high production cost, firms between 53.1% and 68.1% were classified as intermediate production cost, and firms above 68.1% were classified as low production cost manufacturers. As shown in Table 4, the highest tertile, represented by the darkest shade of green, corresponds to low production cost manufacturers. The highest tertile contains almost all top-quartile firms in NAICS 3343 and NAICS 3346. The lowest tertile, represented by the lightest shade of green, corresponds to high production cost manufacturers. The lowest tertile contains most bottom quartile and lower middle quartile firms across every subsector, especially NAICS 3344 and 3345.

As shown in the box plot in Figure 3, the three tertiles display the variations in the spread of gross margin. The low tertile has the smallest range, from the mid 30% to just under 50% gross margin, representing firms with the highest production costs. The middle tertile, centered around 53-60% represents firms with intermediate production costs, and the high tertile spans the largest range, from approximating 70% to nearly 100% gross margin. It is not surprising that the

highest gross margin tertile contains many of the firms in the top revenue quartile. Large scale manufacturers typically benefit from lower input prices, automation and the spreading of fixed costs across large volumes, as described in Besanko et al. (2017). They achieve lower per-unit costs because of their scale, which may explain why they appear in both the top revenue quartile and the highest tertile of gross margin. The opposite is true for smaller manufacturers. Firms in the low quartile, many of which fall into the bottom or lower middle revenue quartiles, face higher input costs and limited ability to spread overhead, all of which translate into high marginal production costs and low gross margins.

Overall, Figure 3 highlights a segmented electronics sector in Ontario. Many small domestic producers cluster in the high production cost tertile; according to Jin et al.'s framework, these firms have the strongest economic incentive to support repair and compete on product longevity rather than rapid replacement. Firms in the middle tertile, or intermediate production cost manufacturers, are expected to initially resist repair before adapting pricing strategies as repair markets mature. On the other hand, the smaller group of highly efficient, high margin producers, which are often multinational electronics firms, have the strongest incentives to discourage repair and follow a volume strategy instead. Taken together, these patterns suggest that Bill 187 may be strategically advantageous for Ontario's smaller manufacturers but may face resistance from high gross margin producers who benefit from frequent product turnover. An example of such resistance is *Apple v. Huseby*, where Apple tried to block an independent Norwegian repair shop from using refurbished iPhone screens by claiming trademark violations (Montello 2020, 172).

Data from Trade Data Online (2025) indicate that Canada's computer and electronic product manufacturing sector operates under a significant and persistent trade deficit, as shown

in Figure 4. Across all subsectors, imports exceed exports by wide margins, as shown in Table 6. Exports are particularly low in computer and peripheral equipment manufacturing and communications equipment manufacturing, where exports amount to only 16.71% and 25% of imports, respectively. Similarly, audio and video equipment manufacturing exports are 23.08% of imports, showing Canada's heavy dependence on imported consumer electronics. Only two subsectors, manufacturing and reproducing magnetic and optical media and navigational, measuring, medical and control instruments manufacturing, show a comparatively high percentage of exports to imports, at over 60% each.

Ontario's dependence on foreign-made electronics provides insight to the central questions of this paper. Since most of the products affected by Bill 187 are imported and designed for global markets, Ontario has limited influence over design decisions related to repairability, durability and planned obsolescence. One potential side effect of Bill 187 is increased compliance costs for importers, which could change product availability. As a result, it can be argued that the effectiveness of legislation targeting repairability relies less on altering foreign manufacturers' design decisions and more on strengthening repair access, affordability and consumer awareness within Ontario.

The trade imbalance calls attention to the importance of the local repair sector. Because the majority of electronics are imported, repairs are the main local economic activity. Thus, increasing repair rates could reclaim value that otherwise would leave the province through imports. Overall, this highlights the fact that Bill 187's most meaningful impacts may be at the post-purchase stage, namely repair.

4.3.2 Overview of Computer and Electronic Product Repair Industry

Canada's repair and maintenance industry is made up of businesses primarily engaged in repairing and maintaining motor vehicles, machinery, equipment and other products. In 2024, there were 79,251 businesses in this sector in Canada, with Ontario containing 33.4% of all employer establishments. The size distribution within Ontario for repairers mirrors that of manufacturers: 59.3% of employer establishments are micro businesses and 40.5% are small, with only 0.2% classified as medium, and virtually no large firms.

The repair business industry is divided into 4 subsectors, with automotive repair and maintenance dominating in both size and revenue. As shown in Figures 3 and 4, this subsector alone includes 47,658 businesses and generates the highest average revenue at $610,000, followed closely by commercial and industrial machinery and equipment. In comparison, electronic and precision equipment repair and maintenance represents a much smaller share of overall repair activity.

Electronic and precision equipment repair shops have several disadvantages relative to automotive repair shops:

- Much smaller market size
- Lower average revenues
- Higher technical complexity

Although they have these disadvantages, electronics and automotive repairers have the same fundamental barriers in common. Bill 187 would remove two significant barriers electronic repairers face: spare parts and repair information.

The strongest empirical evidence on RTR impacts comes from Massachusetts' 2012 automotive right-to-repair law, which required automakers to provide independent repair shops

and consumers with the same diagnostic and repair information previously reserved for authorized dealers. As is the case in Ontario's electronics sector today, this information asymmetry had put small, non-authorized shops at a competitive disadvantage. Kahane (2021) found that the law increased the number of small repair shops by an estimated 134 shops within 6 years and raised the market share of small repair shops by 2.35 to 3.33 percentage points.

Since Ontario's electronics repair sector is made up almost entirely of micro and small firms, Bill 187 is likely to produce similar effects. By removing these barriers, it can be reasonably expected that there will be an increased entry of new electronics repair businesses. In addition, it will lessen operational burdens through eliminating time previously spent on diagnosis and spare parts, increasing the survival rates for existing firms. Over time, these two effects will lead to an overall increase in market share for independent electronics repair shops.

4.3.3 Computer and Electronic Product Consumers

Consumer behaviour plays a central role in determining the effectiveness of right-to-repair legislation. Despite the growing right-to-repair movement, Canadians continue to replace electronic products at high rates. Data shows that over three quarters of Canadians have discarded or replaced a broken device despite the issue being repairable (Innovative Research Group 2019). This disconnect is also reflected in low national repair rates: only 18.6% of Canadian respondents with a broken appliance went the repair route, and of the 81.4% who did not have their product repaired, the majority (61.2%) had not even considered repair as an option (Équiterre 2022, 40). Using data from Statistics Canada, Figure 7 shows the percentage of electronic waste repaired in 2023 is low amongst all categories, ranging from 0% to the highest at 16% for gaming equipment.

Marikyan and Papagiannidis (2024) identify three factors influencing consumers' repair intentions: pro-environmental behaviour, repair-related factors and beliefs about legislation (Marikyan and Papagiannidis 2024, 35). Canadian consumers demonstrate relatively low awareness of right-to-repair, with only a quarter somewhat familiar with the concept (Innovative Research Group 2019). However, support is high both among those who hear about the policy for the first time and those who know details, with 75% of Canadians saying they would support right-to-repair legislation (Innovative Research Group 2019).

Roskladka et al. (2023) have found that convenience of repair and the technical possibility of repair are the two most significant barriers to repair, which is reinforced by Canadian consumers' repair habits. In a 2019 online survey consisted of 1,691 Canadian respondents, among those who discarded and replaced an electronic device, 52% tried to repair overall, 28% tried but it was too expensive and 18% tried and it was not possible.

As found by Jaeger-Erben et al. (2020), consumer practices and routines have an important role. Canadians' view of the repair sector is fairly mixed. While 35% have a positive view of the field, 21.3% of respondents have a negative one and 36.3% a neutral one (Équiterre 2022, 41). A majority of the public blame planned obsolescence as a key barrier to repair, and a minority believe that manufacturers facilitate the repair of their appliances (Équiterre 2022, 46). Amongst possible measures that could be implemented by authorities, lower repair prices, higher product quality, and replacement parts that can be accessed over at least 10 years were the top three supported measures (Équiterre 2022, 46).

Taken together, these findings show that Bill 187 will make repair more accessible, but barriers like low consumer awareness, inconvenience and planned obsolescence will remain. Low awareness of repair options, mixed trust in the repair sector and the different types of

planned obsolescence mean that many Ontarians will likely continue to replace devices at a high rate. Broader policy and educational measures that normalize repair, improve confidence in independent repair services and address product longevity would help increase repair rates. Without addressing these issues, the possibility of repair may increase without meaningfully changing the likelihood that consumers will choose repair.

4.3.4 Possible Environmental Implications

As shown in Figure 8, electronics, electrical and electronic equipment was the fourth highest product type of discarded plastic in Canada in 2021. Existing research suggests that longer product use and improved e-waste collection are the two most impactful levers in reducing landfilled electronic waste (Nagase and Uehara 2024, 184). Nagase and Uehara's (2024) case study in the UK of the potential impact of new 'Right to Repair' rules on electrical and electronic equipment (EEE) waste found that compared with the business-as-usual scenario, one year longer EEE use and 10% more of WEEE collection could reduce landfilled WEEE by 14.05% of monitoring and control instruments.

Although these calculations are specific to the UK, Bill 187 affects the same underlying mechanism required to decrease product turnover. The possible environmental implications of Bill 187 are closely tied to how effectively it extends the lifespan of electronic products and alters end-of-life waste flows in Ontario. Given Ontario's high consumption of imported electronics and limited domestic manufacturing, the most meaningful benefits of Bill 187 will arise not from product design changes but from increasing repair rates and slowing the rate of device replacement.

Canadian data further highlights the importance of lifespan extension. Habib et al. (2023) estimate that e-waste generation per capita in Canada has nearly tripled over the past two

decades, rising from 8.3kg in 2000 to 25.3kg in 2020, and is projected to reach 31.5kg by 2030. The study highlights that Canada's growing stock of consumer electronics is the primary driver of rising e-waste volumes.

Taken together, the findings from Nagase and Uehara (2024) and Habib et al. (2023) suggest that Bill 187's most significant environmental benefit will come from delaying device disposal. Even marginal increases in average device lifespan can compound into substantial reductions in provincial e-waste generation over time. While the exact magnitude will depend on manufacturer compliance and consumer behaviour, Bill 187 positions Ontario to slow the upward trajectory of e-waste accumulation.

5 Discussion

This paper set out to evaluate the potential implications of Bill 187 on Ontario's electronics industry, consumers and environmental outcomes. Altogether, the analysis shows that the bill represents an important but incomplete step toward expanding repair access in Ontario. By mandating the availability of repair manuals, diagnostic tools and spare parts, Bill 187 removes two of the most prevalent access barriers identified in the literature. These changes are especially meaningful considering Ontario relies heavily on imported electronics, meaning that local economic activity is more concentrated in repair than manufacturing. As a result, Bill 187 has the potential to expand Ontario's independent repair sector and reclaim economic value that currently leaves the province through the purchase of replacement devices.

However, the results also highlight substantial limitations in Bill 187's ability to address deeper structural barriers to repair. Competitive obstacles, such as high repair prices relative to replacement, planned obsolescence and convenience and trust remain largely untouched. These gaps are not minor, as international examples show that RTR frameworks require more broad

interventions. In the European Union, measures like long-term parts availability requirements, anti-obsolescence rules and repairability indexes demonstrate a more committed and robust plan for tackling these barriers. Without such measures, manufacturers with strong incentives to maintain rapid product turnover, particularly large global multinational companies, may continue to design products that undermine repairability.

The paper also calls attention to the important role of consumer behaviour. Low awareness of repair options, mixed trust in independent repairers and perceived inconvenience are some of the factors that suggest increased access to repairs may not be sufficient to increase the currently low repair rates. Without policies that target consumer education or psychological drivers of replacement, replacement could remain the default choice for many Ontarians.

Environmental implications are drawn from research showing that extending product lifespans is one of the most effective strategies for reducing electronic waste. Bill 187's largest environmental benefit will come from delaying device disposal, as even marginal increases in average device lifespan compound into substantial reductions over time.

All in all, the findings point to the conclusion that Bill 187 is a meaningful first step that will aid repair in Ontario's electronics sector, but also leaves significant policy gaps. While Bill 187 will make repair more technically possible for consumers, stronger trust and shifts in consumer behaviour are still needed for Ontario's repair sector to grow and for small Ontario manufacturers to gain a competitive advantage. Without addressing affordability, planned obsolescence and the factors that make repair untrustworthy or inconvenient, the bill risks giving consumers new rights that are not exercised in practice. Adding clearer requirements for long-term parts availability, anti-obsolescence measures, and policies that will improve consumer awareness would align Ontario more closely with international RTR legislation.

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Tables

Table 1: Overview of Repair Barriers Mentioned in Lopez-Bermudez and Vence (2025)

Type of Barrier	Barriers to repair identified in the Literature	Description
Access barriers ➔ Impede repair altogether, even when it is possible and desired by the consumer	Legal obstacles	Include Extended Producer Responsibility (EPR) and e-waste legislation, legal guarantees, and intellectual property (IP) law.
	Spare parts	Limiting access to spare parts can prevent repairs from taking place and channel business through authorized repair services, potentially increasing the cost.
	Information	Manufacturers are not required to provide manuals and information on how to repair their products, and unauthorized distribution of such information can be considered an infringement of copyright law.
Competitive barriers ➔ Result from market structures that make repair less attractive and therefore less likely to occur	Prices	Identified as a main barrier to repair, factors that determine the relative prices of repair will affect either the price of repair or the price of new products.
	Material Obsolescence	Reducing the quality of materials and increasing the use of plastics, which often cannot be repaired without it being noticeable. A form of material obsolescence for electronics is programmed obsolescence, e.g. the designing of circuits that redirect energy through

		certain nodes after the legal guarantee expires.
	Functional Obsolescence	Tied to innovation and legal aspects, refers to products being phased out because the technology has changed in such a way that the product is either useless or not as effective as it was.
	Psychological Obsolescence	Refers to obsolescence in the mind of the consumer, rather than in the material worlds. Factors such as advertising and marketing encourage people to replace products in the search for novelty, and the perception of something as obsolete prevents users from attempting to repair it.
	Convenience and trust	General uncertainty about the quality of repairers' work, and distrust regarding repaired and second-hand products.

Table 2: Repair Barriers and Bill 187 Assessment

Barriers to Repair Identified in the Literature	Bill 187 Assessment
Legal obstacles	Not addressed
Spare parts	Addressed
Information	Addressed
Prices	Partially addressed
Material Obsolescence	Not addressed
Functional Obsolescence	Not addressed

Psychological Obsolescence	Not addressed
Convenience and trust	Not addressed

Table 3: Manufacturers' Strategy Based on Production Cost

Production cost	Manufacturer actions
Low - "no-repair"	Manufacturer chooses: ➔ Charge an exorbitantly high repair price ➔ Relatively low new product price to churn out new products in volume Focus on mitigating decreased demand by hampering repair – new units are cheap enough that repair is discouraged and consumers just buy more devices
Intermediate - "partial-repair"	Manufacturer chooses: ➔ Follow the *volume strategy* initially but as the independent repair cost continues to decline and falls below a certain point, switches the a *margin strategy* by raising the new product price and offer free repair
High - "full-repair"	Manufacturer chooses: ➔ Free repair ➔ High new product price that includes expected repair cost Since products are expensive to make, the firm wants them to last and gives free repair/warranty to raise the value and justify the high price.

Table 4: Financial Performance of NAICS 334 Manufacturing Subsectors

NAICS code	Manufacturing sector	Quartile	Total revenue	Cost of sales (direct expenses)	Net profit/loss	Gross margin
3344	Semiconductor and other electronic component	Lower middle (25%)	$245,500	$82,000	-$185,500	36.4%

	manufacturing					
3342	Communications equipment manufacturing	Bottom quartile (25%)	$42,700	$8,300	$1,900	43.2%
3341	Computer and peripheral equipment manufacturing	Lower middle (25%)	$125,600	$34,200	$11,400	44.7%
3345	Navigational, measuring, medical and control instruments manufacturing	Lower middle (25%)	$299,000	$92,400	-$115,000	46.2%
3343	Audio and video equipment manufacturing	Bottom quartile (25%)	$44,300	$4,400	$18,400	47.1%
3341	Computer and peripheral equipment manufacturing	Bottom quartile (25%)	$52,800	$12,500	-$3,300	48.3%
3346	Manufacturing and reproducing magnetic and optical media	Lower middle (25%)	$101,300	$15,200	-$37,300	49.0%
3344	Semiconductor and other electronic component manufacturing	Upper middle (25%)	$1,081,100	$553,300	-$192,000	49.6%
3342	Communications equipment manufacturing	Lower middle (25%)	$112,500	$40,600	-$22,800	53.1%
3345	Navigational, measuring, medical and control instruments manufacturing	Upper middle (25%)	$939,500	$375,300	-$265,400	53.5%
3345	Navigational, measuring, medical and control instruments manufacturing	Bottom quartile (25%)	$78,900	$18,600	-$54,500	55.0%
3343	Audio and video equipment manufacturing	Lower middle (25%)	$78,400	$12,700	$13,300	56.3%
3342	Communications equipment manufacturing	Upper middle (25%)	$359,300	$119,800	-$8,900	58.4%
3341	Computer and peripheral equipment manufacturing	Upper middle (25%)	$318,800	$121,200	-$18,200	58.9%
3344	Semiconductor and other electronic component manufacturing	Top quartile (25%)	$3,099,200	$1,580,400	-$63,700	67.7%
3344	Semiconductor and other electronic component manufacturing	Bottom quartile (25%)	$78,300	$19,500	-$75,100	68.1%
3346	Manufacturing and reproducing magnetic and optical media	Upper middle (25%)	$172,400	$10,300	-$174,400	68.8%
3345	Navigational, measuring, medical and control instruments manufacturing	Top quartile (25%)	$3,048,800	$1,517,400	-$84,000	68.9%
3346	Manufacturing and reproducing magnetic and optical media	Bottom quartile (25%)	$48,300	$6,200	$10,200	71.8%
3343	Audio and video equipment manufacturing	Upper middle (25%)	$201,800	$69,800	$40,400	79.7%
3342	Communications equipment manufacturing	Top quartile (25%)	$2,119,700	$1,024,400	-$33,800	87.0%

3341	Computer and peripheral equipment manufacturing	Top quartile (25%)	$1,765,800	$887,500	-$69,900	89.6%
3343	Audio and video equipment manufacturing	Top quartile (25%)	$1,413,600	$610,600	$140,200	95.7%
3346	Manufacturing and reproducing magnetic and optical media	Top quartile (25%)	$745,400	$319,500	-$79,800	98.4%

Source: Canadian Industry Statistics (2025)

Table 5: Tertile Cutoffs for Gross Margin

```
Tertile cutoffs for 'Gross margin':
  Lower cutoff (1/3 quantile): 53.099999999999994
  Upper cutoff (2/3 quantile): 68.1

  Tertile 1 (low): 8 rows
  Tertile 2 (mid): 8 rows
  Tertile 3 (high): 8 rows
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Source: Code Output From Table 3 Values

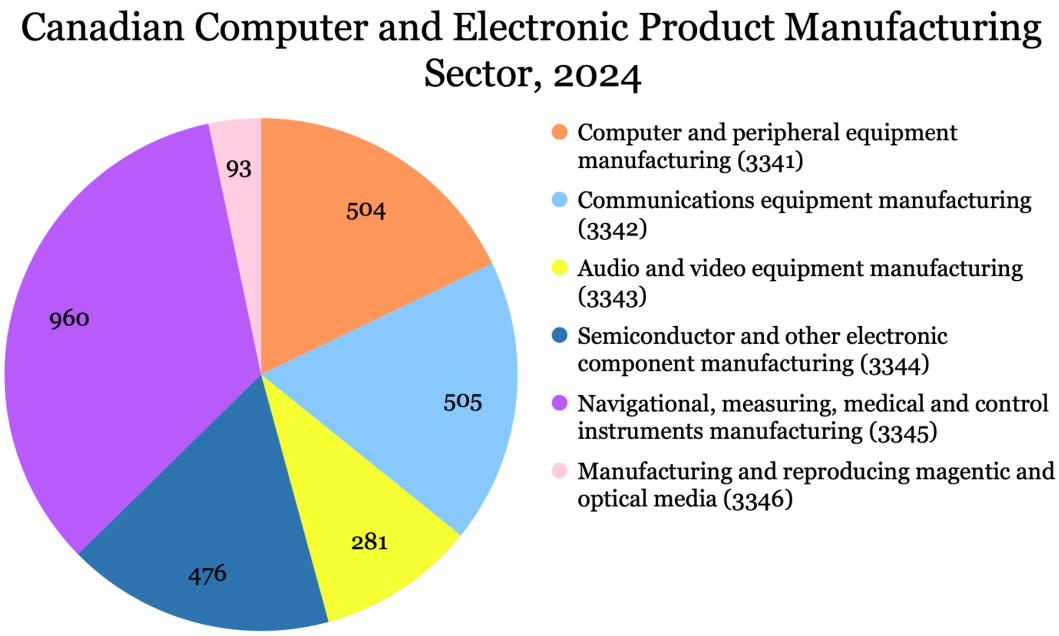
Table 6: Manufacturing Subsector Exports as a Percentage of Imports

Manufacturing sector	Exports as a percentage of imports
Audio and video equipment manufacturing	23.08%
Communications equipment manufacturing	25.00%
Computer and peripheral equipment manufacturing	16.71%
Manufacturing and reproducing magnetic and optical media	66.34%
Navigational, measuring, medical and control instruments manufacturing	60.92%
Semiconductor and other electronic component manufacturing	35.62%

Source: Trade Data Online (2025)

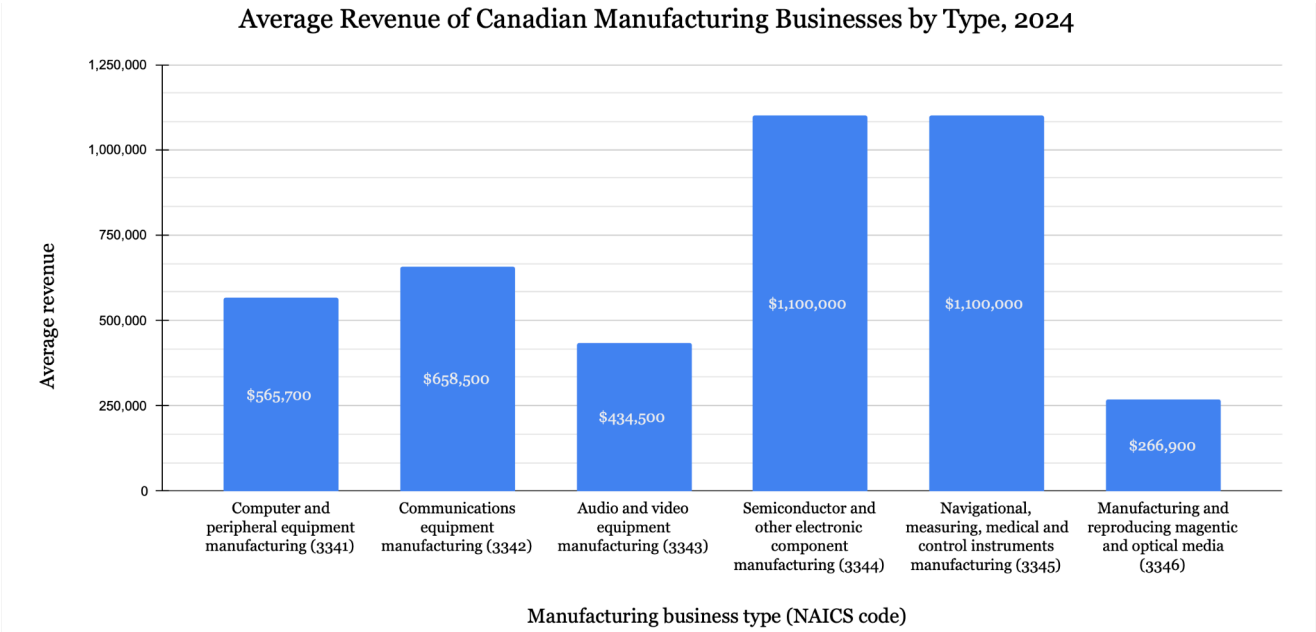
Figures

Figure 1: Breakdown of Computer and Electronic Product Manufacturing Businesses in Canada



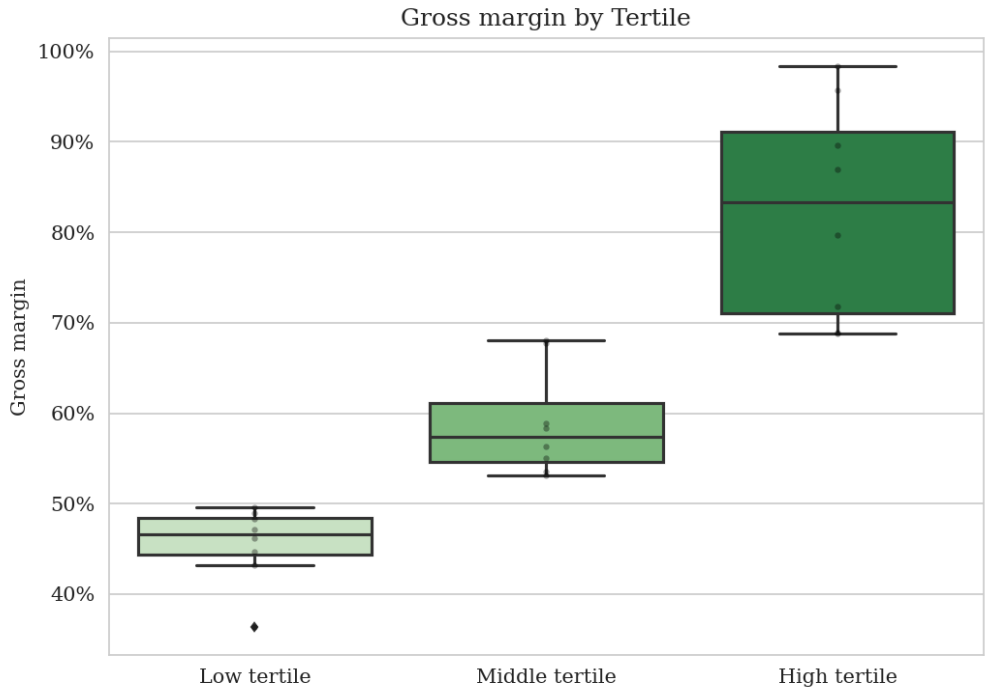
Source: Canadian Industry Statistics (2025)

Figure 2: Average Revenue of Canadian Manufacturing Businesses by Subsector



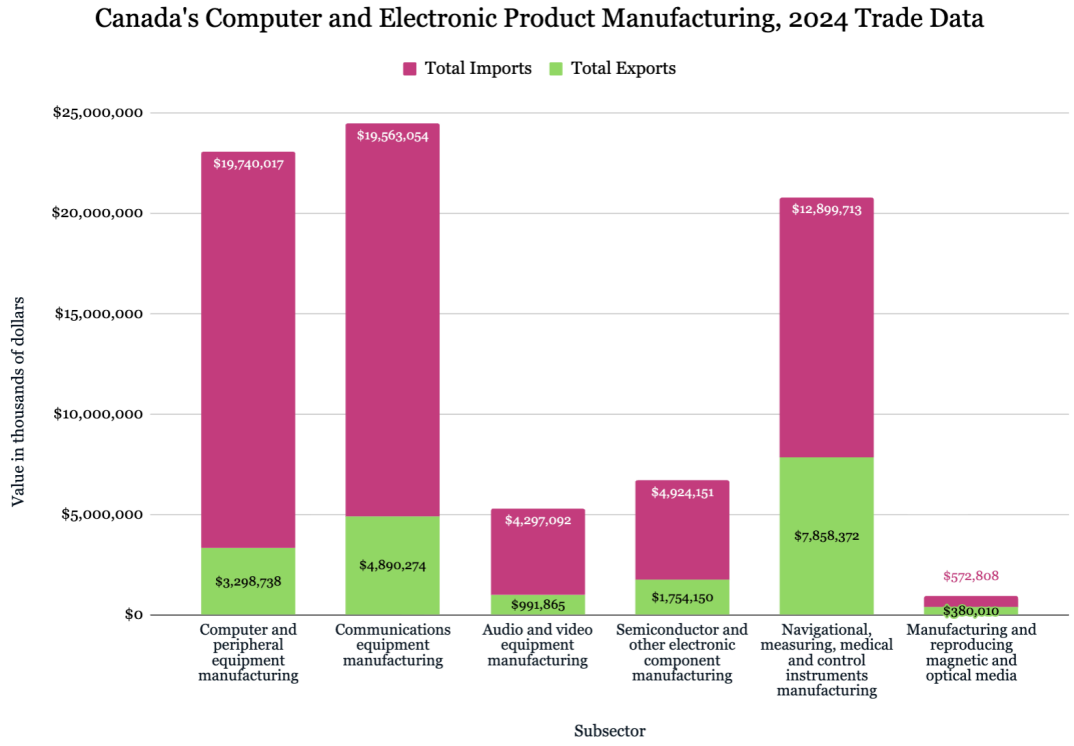
Source: Canadian Industry Statistics (2025)

Figure 3: Box Plot of Gross Margins for Canadian Computer and Electronics Manufacturers



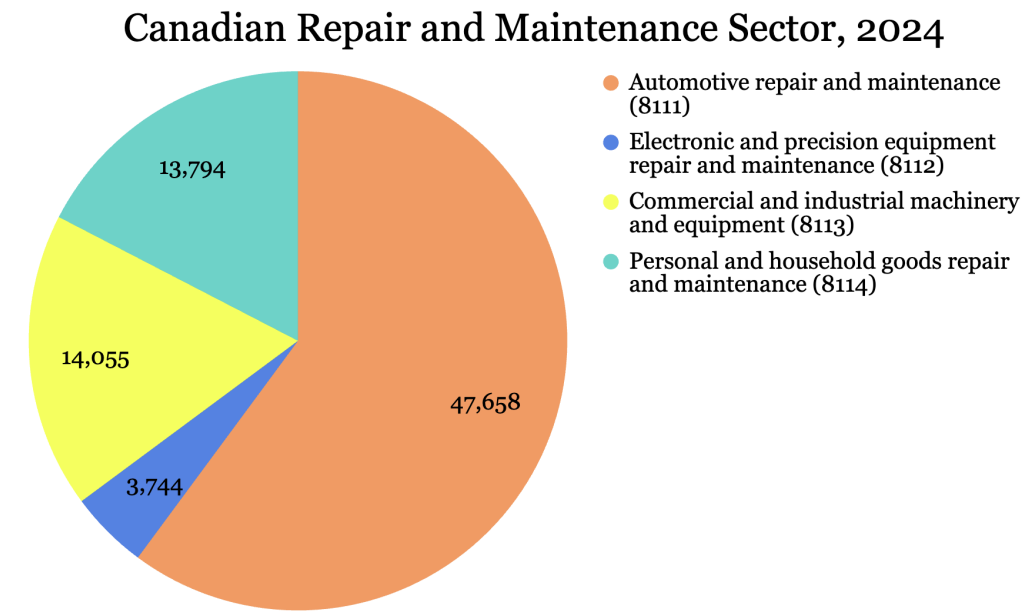
Source: Canadian Industry Statistics (2025)

Figure 4: Imports and Exports of Canada's Computer and Electronic Product Manufacturers



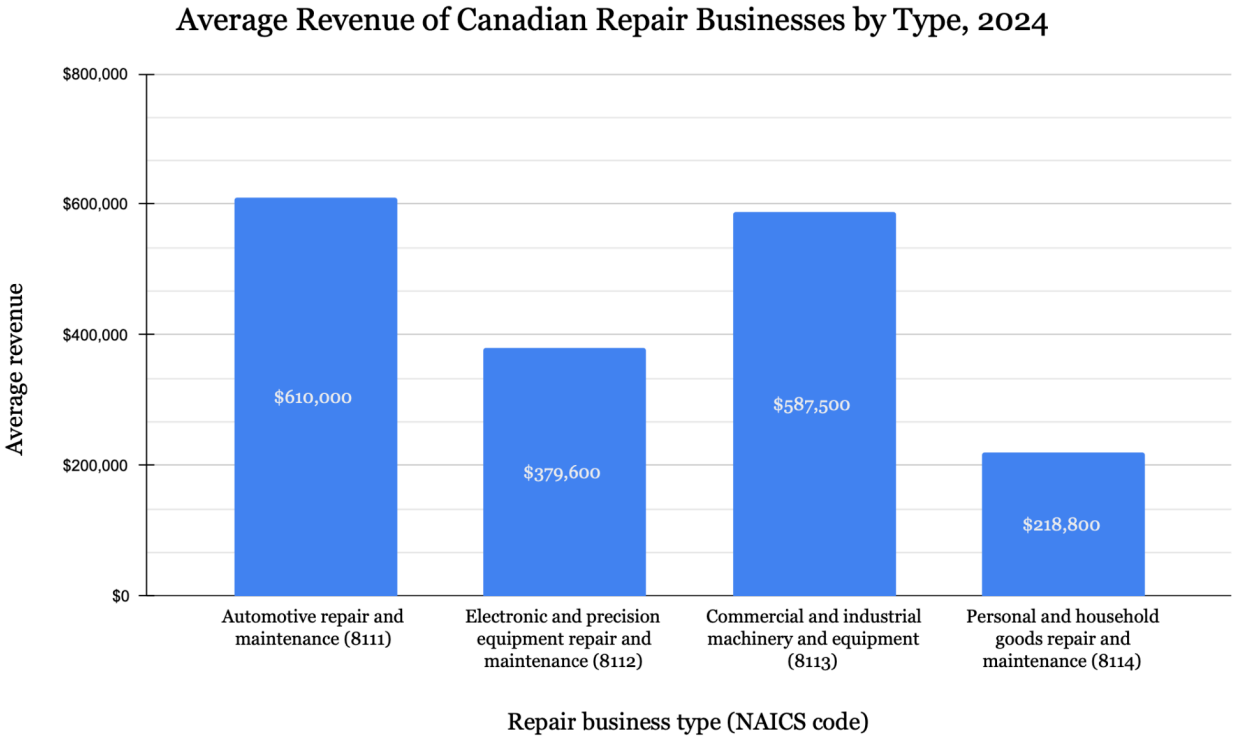
Source: Trade Data Online (2025)

Figure 5: Breakdown of Repair and Maintenance Businesses in Canada



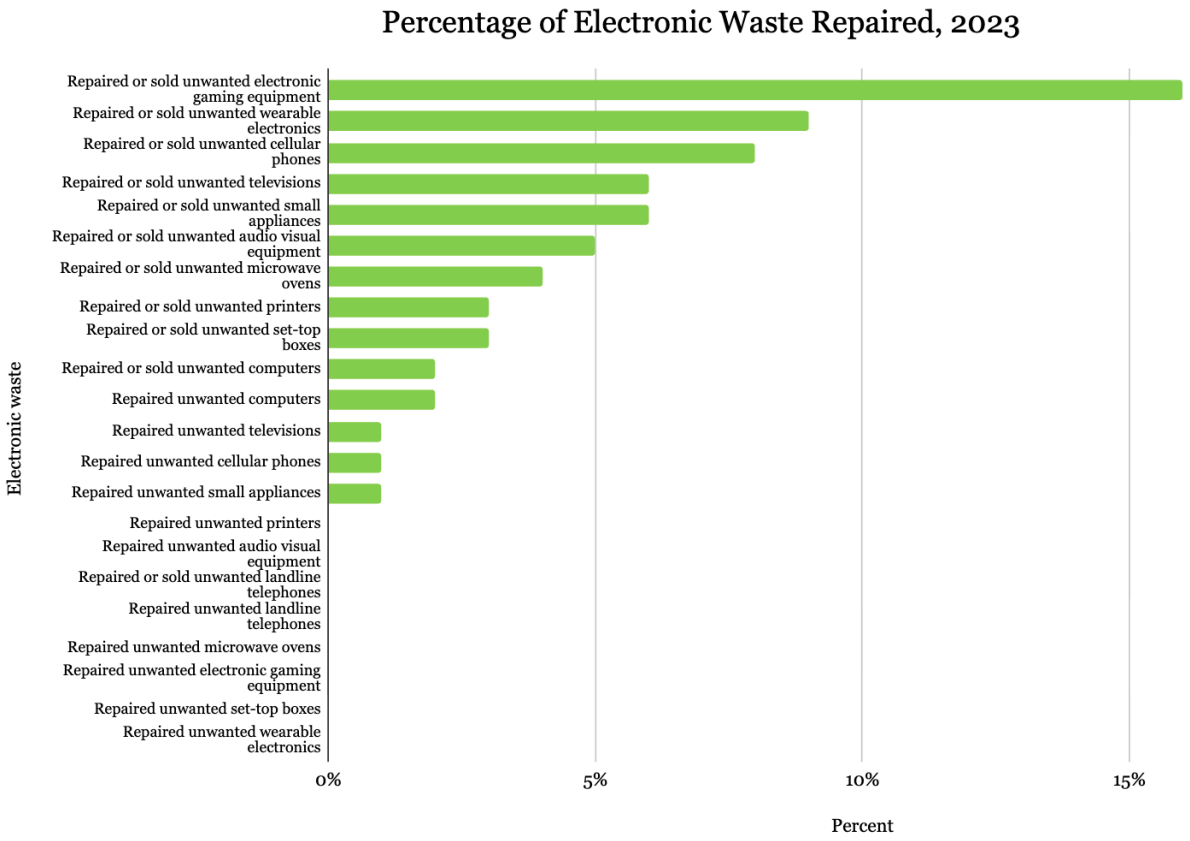
Source: Canadian Industry Statistics (2025)

Figure 6: Average Revenue of Canadian Repair Businesses by Subsector



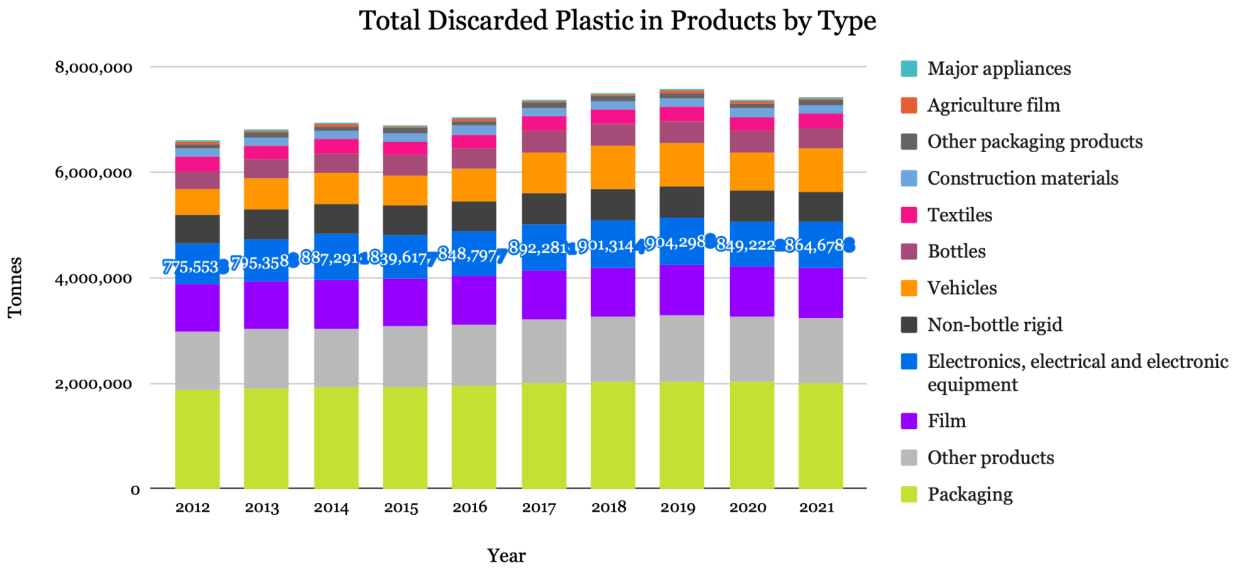
Source: Canadian Industry Statistics (2025)

Figure 7: Percentage of Electronic Waste Repaired by Product Type



Source: Statistics Canada (2025)

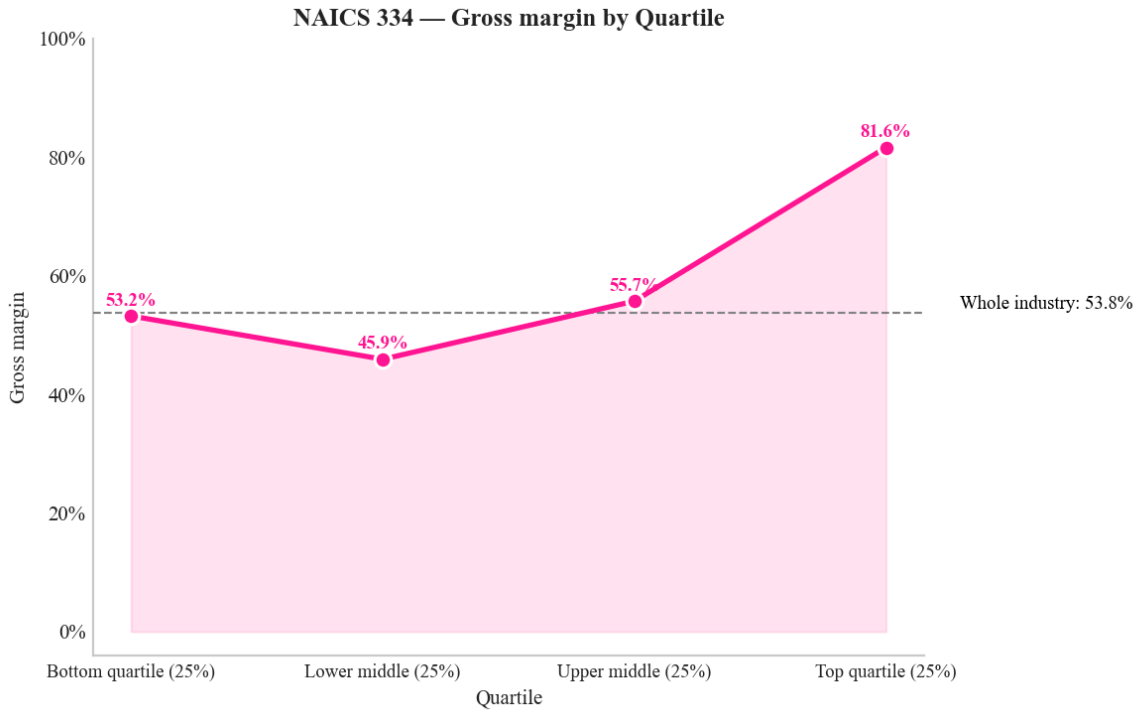
Figure 8: Total Discarded Plastic in Products by Type

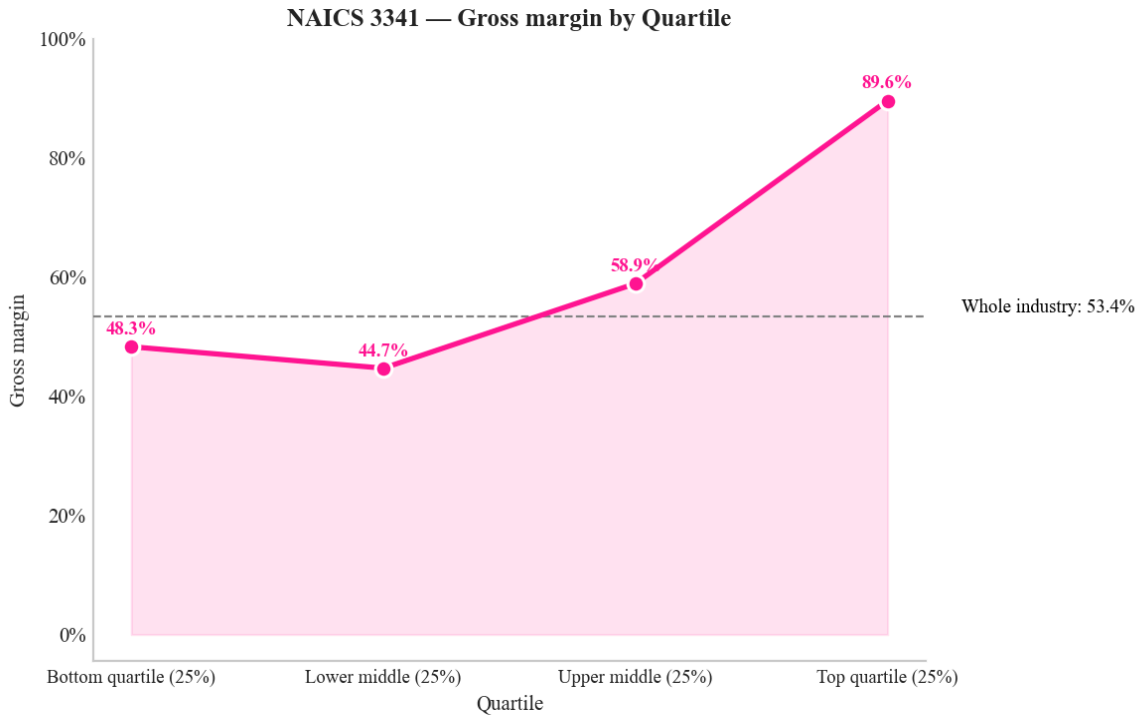


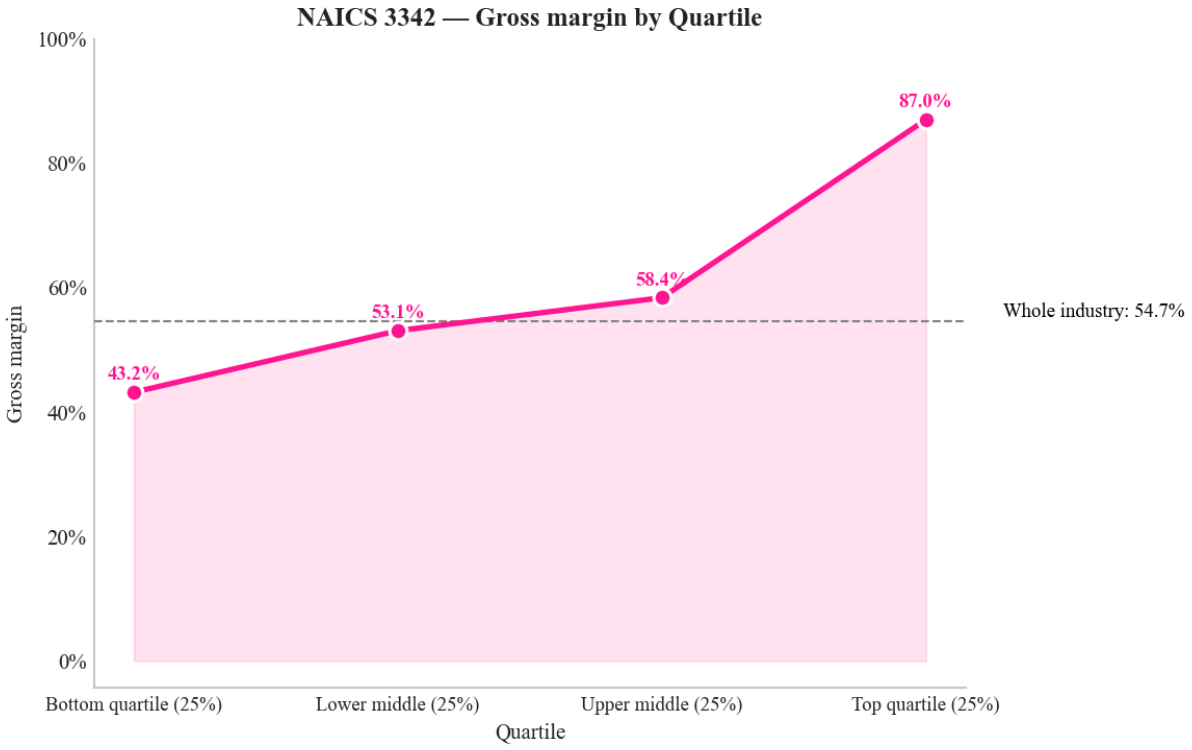
Source: Statistics Canada (2025)

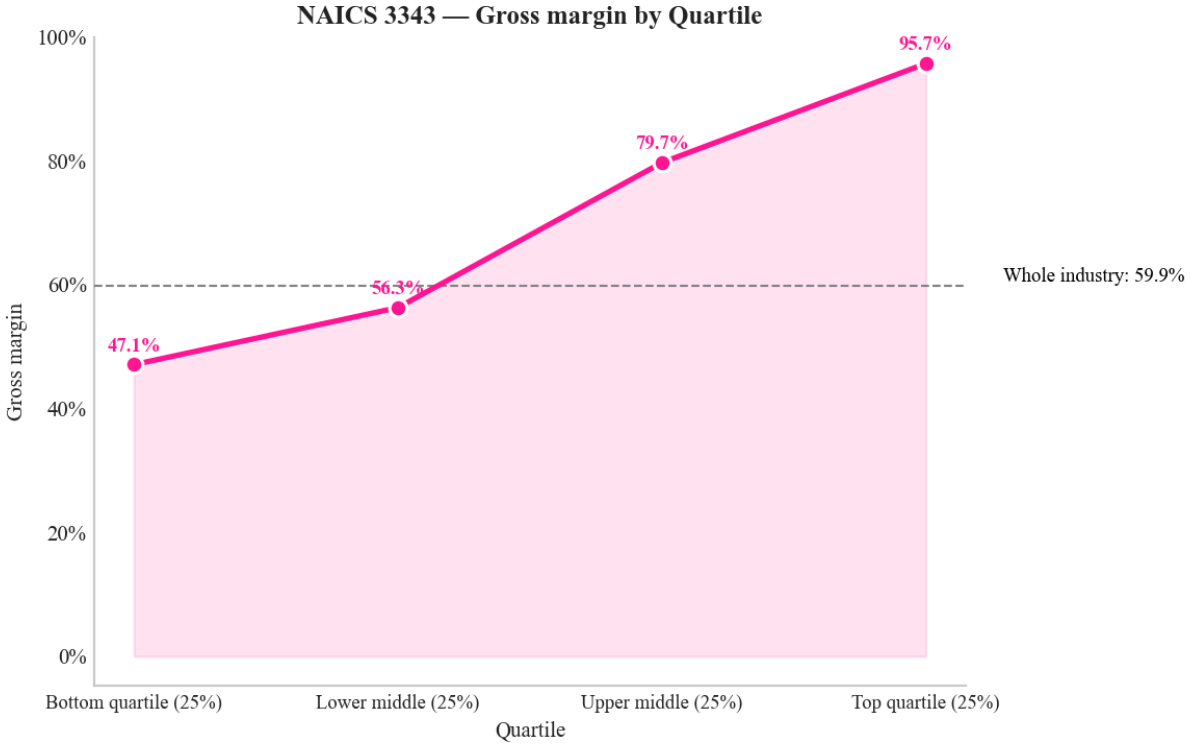
7 Appendix – Gross Margin Graphs

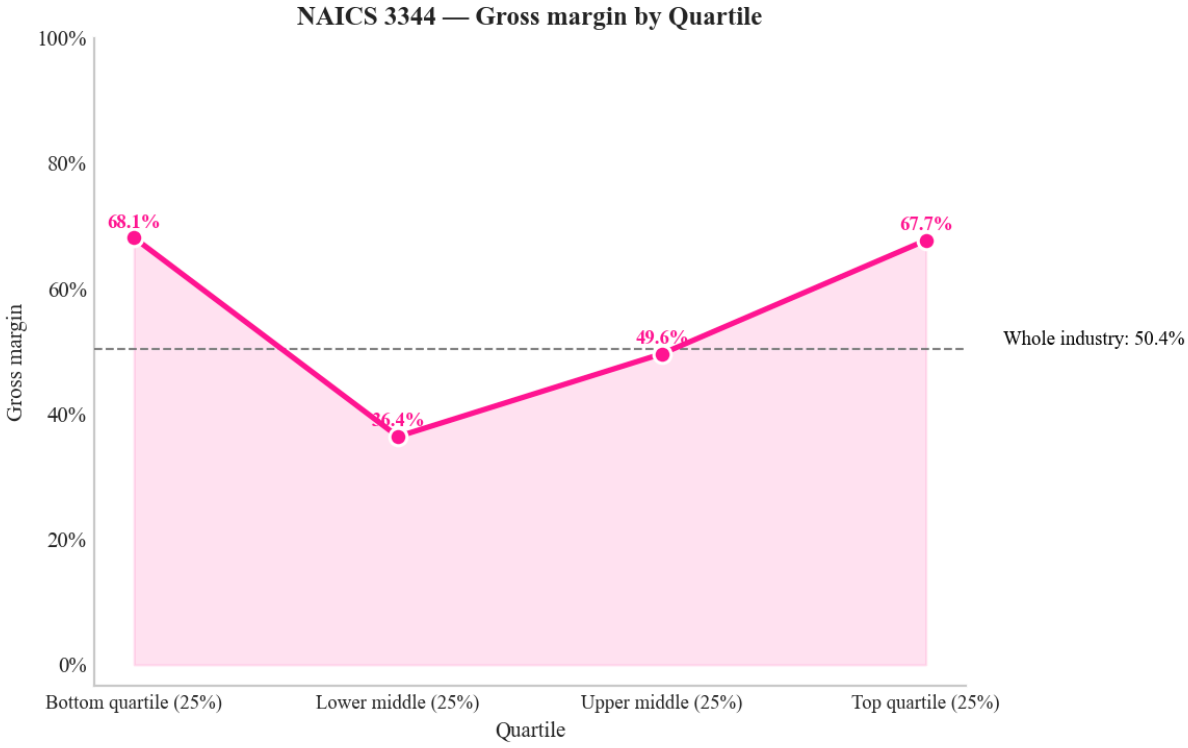
Source: Canadian Industry Statistics

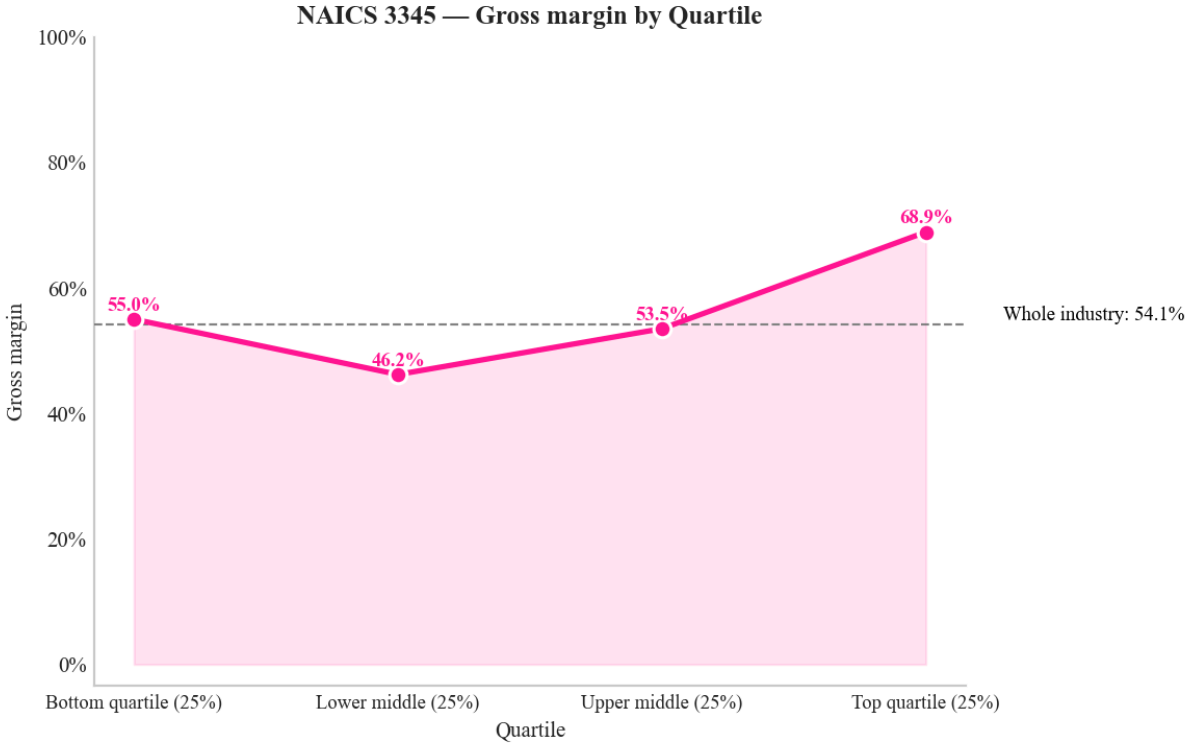


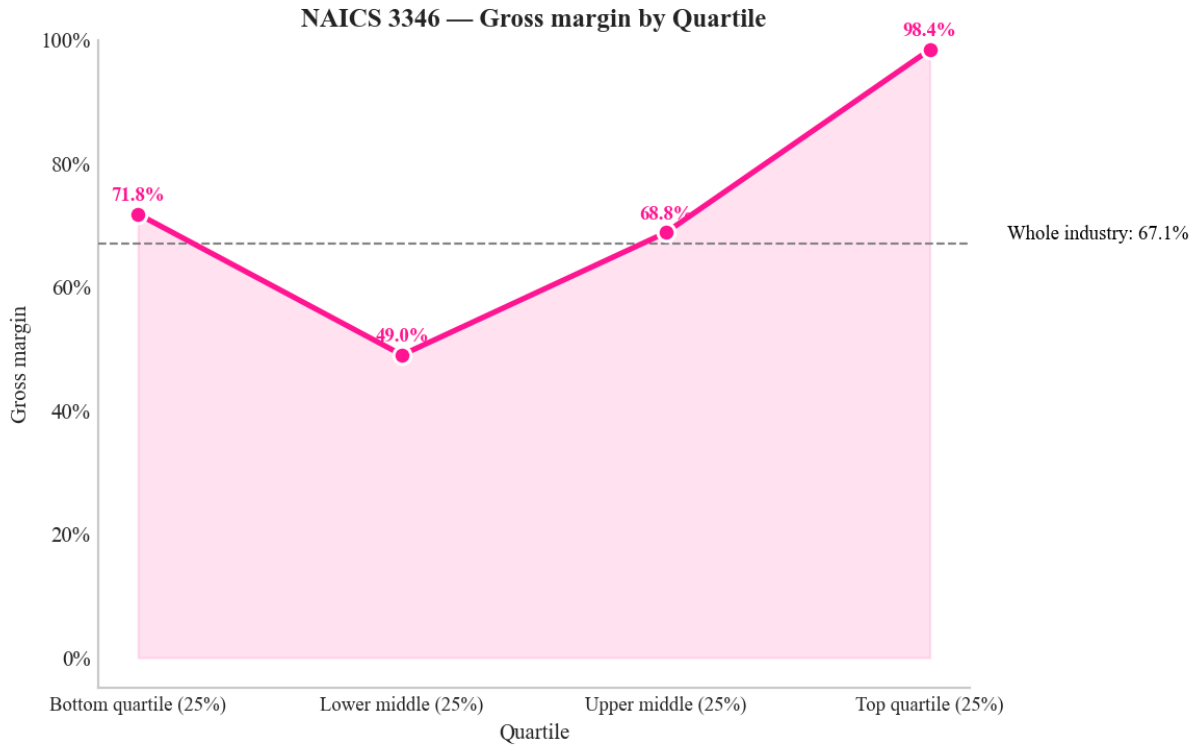


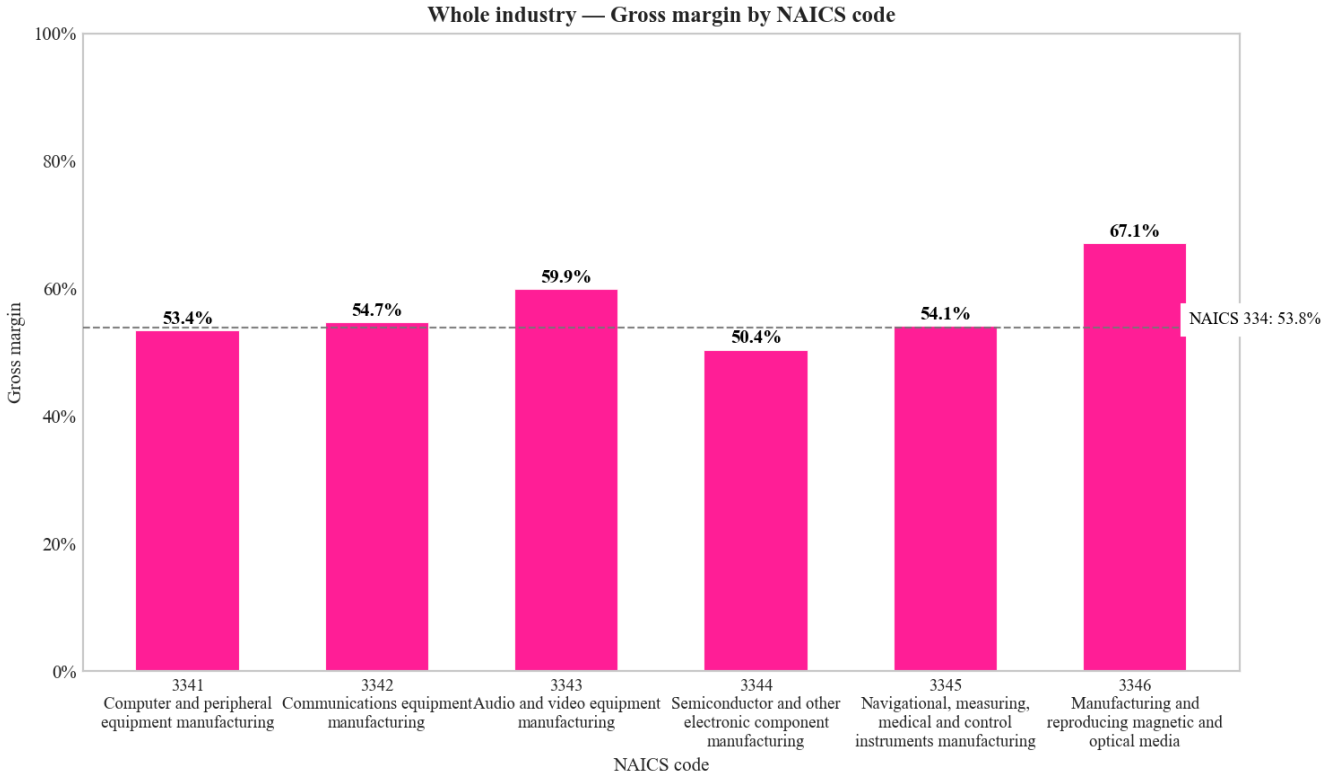

NAICS 3342 — Gross margin by Quartile
100%
80%
60%
40%
20%
0%
Gross margin
43.2%
53.1%
58.4%
87.0%
Whole industry: 54.7%
Bottom quartile (25%)
Lower middle (25%)
Upper middle (25%)
Top quartile (25%)
Quartile


NAICS 3343 — Gross margin by Quartile
100%
80%
60%
40%
20%
0%
Gross margin
47.1%
56.3%
79.7%
95.7%
Whole industry: 59.9%
Bottom quartile (25%)
Lower middle (25%)
Upper middle (25%)
Top quartile (25%)
Quartile


NAICS 3344 — Gross margin by Quartile
100%
80%
60%
40%
20%
0%
Gross margin
68.1%
36.4%
49.6%
67.7%
Whole industry: 50.4%
Bottom quartile (25%)
Lower middle (25%)
Upper middle (25%)
Top quartile (25%)
Quartile


NAICS 3345 — Gross margin by Quartile
100%
80%
60%
40%
20%
0%
Gross margin
55.0%
46.2%
53.5%
68.9%
Whole industry: 54.1%
Bottom quartile (25%)
Lower middle (25%)
Upper middle (25%)
Top quartile (25%)
Quartile


NAICS 3346 — Gross margin by Quartile
100%
80%
60%
40%
20%
0%
Gross margin
71.8%
49.0%
68.8%
98.4%
Whole industry: 67.1%
Bottom quartile (25%)
Lower middle (25%)
Upper middle (25%)
Top quartile (25%)
Quartile


Whole industry — Gross margin by NAICS code
100%
80%
60%
40%
20%
0%
Gross margin
53.4%
54.7%
59.9%
50.4%
54.1%
67.1%
NAICS 334: 53.8%
3341
Computer and peripheral equipment manufacturing
3342
Communications equipment manufacturing
3343
Audio and video equipment manufacturing
3344
Semiconductor and other electronic component manufacturing
3345
Navigational, measuring, medical and control instruments manufacturing
3346
Manufacturing and reproducing magnetic and optical media
NAICS code